

KOCHI, India

WMO: 433530

Lat: 9.946N

Lon: 76.272E

Elev: 2

StdP: 101.30

Time Zone: 5.50 (IND)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.8	23.3	16.7	11.9	28.3	17.7	12.7	27.8	9.7	25.6	7.9	26.6	1.4	320	0.356

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	6.2	33.2	26.8	32.7	26.5	32.3	26.3	28.0	31.7	27.6	31.3	27.3	31.0	4.9	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
27.0	22.6	30.4	26.6	22.1	30.1	26.2	21.6	29.8	89.5	31.8	87.8	31.5	86.4	31.3	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.4	6.4	5.8	20.4	35.4	1.3	2.0	19.5	36.9	18.8	38.0	18.1	39.1	17.2	40.6		
			18.4	29.2	1.1	0.6	17.5	29.7	16.9	30.0	16.2	30.3	15.4	30.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	27.8	27.5	28.2	29.1	29.3	28.9	27.2	26.7	26.8	27.2	27.4	27.7	27.6
	DBStd	1.30	0.78	0.86	0.86	1.02	1.28	1.33	1.08	1.07	0.88	0.89	0.77	0.88
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6502	544	510	592	579	587	517	519	521	516	540	531	547
	CDD18.3	3461	286	277	334	329	328	267	261	262	266	282	281	288
	CDH23.3	38114	3054	3201	4249	4280	4166	2674	2351	2414	2668	2907	3027	3123
	CDH26.7	13485	1070	1212	1866	2001	1839	783	542	536	691	868	997	1080
Wind	WSAvg	2.3	2.1	2.2	2.3	2.5	2.5	2.3	2.4	2.5	2.5	2.2	1.8	1.9
Precipitation	PrecAvg	3242	31	21	37	115	344	674	759	414	321	289	183	54
	PrecMax	4736	169	170	139	292	850	1134	1454	646	588	513	476	276
	PrecMin	2263	0	0	0	36	25	219	457	182	74	89	17	0
	PrecStd	518	49	38	36	56	279	260	246	140	155	117	122	81
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.4	33.1	33.5	33.9	33.7	32.2	31.5	31.1	31.4	31.8	32.2	32.5
		MCWB	23.7	24.5	26.5	27.3	27.5	27.0	26.9	26.4	26.3	26.1	25.5	24.7
	2%	DB	31.7	32.2	32.8	33.2	33.0	31.5	30.7	30.3	30.7	31.1	31.6	31.7
		MCWB	23.8	24.5	26.3	27.1	27.2	26.8	26.4	26.0	25.9	26.0	25.5	24.6
	5%	DB	31.2	31.7	32.3	32.7	32.5	30.9	30.0	29.8	30.2	30.6	31.1	31.2
		MCWB	23.7	24.5	26.2	27.0	27.1	26.6	26.1	25.8	25.8	25.9	25.5	24.5
	10%	DB	30.7	31.1	31.8	32.2	31.9	30.1	29.3	29.2	29.7	30.1	30.6	30.7
		MCWB	23.7	24.5	26.0	26.9	27.0	26.3	25.9	25.6	25.6	25.7	25.4	24.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.4	26.8	28.0	28.5	28.6	27.9	27.5	27.1	27.2	27.2	27.2	26.9
		MCDB	29.9	31.0	31.7	32.2	32.1	31.0	30.4	30.0	30.2	30.4	30.8	30.8
	2%	WB	25.7	26.2	27.2	27.9	28.0	27.4	26.9	26.5	26.6	26.7	26.7	26.2
		MCDB	29.5	30.2	31.4	31.9	31.7	30.4	29.8	29.4	29.7	30.0	30.3	30.0
	5%	WB	25.3	25.8	26.9	27.5	27.6	27.0	26.5	26.2	26.2	26.2	26.2	25.7
		MCDB	29.1	29.7	31.1	31.6	31.3	30.0	29.2	28.9	29.2	29.5	29.7	29.4
	10%	WB	24.9	25.5	26.6	27.2	27.2	26.6	26.1	25.8	25.9	26.0	25.9	25.4
		MCDB	28.8	29.4	30.7	31.3	30.9	29.4	28.6	28.4	28.7	29.2	29.3	29.0
Mean Daily Temperature Range	5% DB	MDBR	7.0	6.5	5.9	6.2	5.7	4.9	4.7	4.5	5.0	5.5	6.1	6.8
		MCDBR	7.4	6.9	6.3	6.4	6.1	5.9	5.6	5.2	5.6	6.1	6.6	7.1
		MCWBR	2.9	2.6	2.4	2.8	2.5	2.5	2.5	2.1	2.2	2.3	2.6	2.8
	5% WB	MCDBR	6.5	6.2	5.9	6.2	6.0	5.4	5.4	5.0	5.2	5.8	6.2	6.4
		MCWBR	3.0	2.5	2.6	3.0	2.8	2.7	2.9	2.4	2.4	2.5	2.8	2.8
Clear-Sky Solar Irradiance	taub	0.502	0.536	0.596	0.601	0.571	0.533	0.554	0.523	0.483	0.537	0.543	0.502	
	taud	2.108	2.020	1.893	1.902	1.991	2.105	2.030	2.132	2.263	2.079	2.040	2.139	
	Ebn at Noon	808	795	754	742	749	768	756	792	833	784	766	799	
	Edh at Noon	159	179	206	202	181	159	172	159	140	166	169	151	
All-Sky Solar Radiation	RadAvg	5.60	6.19	6.34	5.99	5.25	4.26	4.28	4.69	5.21	4.93	4.72	5.01	
	RadStd	0.20	0.21	0.26	0.22	0.38	0.39	0.38	0.40	0.46	0.29	0.27	0.24	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (4 neighbors)	+0.40	N/A	N/A	+0.70	+0.48	+0.43	N/A	N/A	N/A	N/A

Nomenclature: See separate page